Yuhui Wang

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SUMMARY

PhD student in Computer Science, Stony Brook University, USA. Specializing in LLM post-training, safety alignment, RAG, and machine learning. 10 peer-reviewed papers (ICLR, S&P), $5K fellowship awardee. Available for Summer 2026 (May to Sept) internship. Expt Grad in 2029.

EDUCATION

Stony Brook University | Computer Science | PhD Student | NY, USA 2024.09 - Present
• Advisor: Prof. Ting Wang • Research Interest: LLM Post-Training, RLHF, Safety Alignment, LLM Security

Shanghai Jiao Tong University | Industrial Engineering | Master's Degree | Shanghai, China 2021.09 - 2024.03
• Advisor: Prof. Di Wang • Research Interest: Multi-dimensional Time-series Classification & Prediction

Zhejiang University | Industrial Engineering | Bachelor's Degree | Hangzhou, China 2017.09 - 2021.06

PROJECT EXPERIENCE

LLM Alignment Improvment | Project Leader | 3 Papers Under Review at ICLR 2024.09 - Present

• Mitigated **fine-tuning jailbreak attacks** by proposing a **self-destructive** defense with a custom loss coupling benign and harmful gradients, implemented via **adapting HuggingFace Trainer**; enable the model to retain **~5%** initial harmful metric or be **fully destroyed** post-attack
• Suppressed retrieval shortcuts in **reinforcement learning** to strengthen model reasoning and ensure **faithful** outputs; improved CoT robustness by **47.8%** and pass@1 accuracy by **5.8%** across diverse benchmarks, implemented through customized **adaptation of the veRL trainer**

GraphRAG Data Poisoning | Project Leader | 1 IEEE S&P Paper 2024.09 - 2025.02

• As the **first data poisoning study on GraphRAG** developed by Microsoft, we proposed GragPoison, a poisoning attack targeting GraphRAG by leveraging its unique relation and entity retrieval mechanism to **amplify the poisoning effect**. Achieved up to **98%** attack success with **<68%** poisoning text on medical, cybersecurity, and commonsense datasets across multiple GraphRAG variants compared with benchmarks

Automatic Close Source Reasoning Model Jailbreak & Defense | 1 ICLR Paper, 1 Paper Under Review at ICLR 2024.09 - Present

• Participated in developing AutoRAN, an **automated** jailbreak attack framework that uses a red-team reasoning model to simulate and refine prompts via target models' intermediate reasoning steps. Achieved **~100%** success rates on GPT-o3/o4-mini and Gemini-2.5-Flash
• **Benchmarked** proposed jailbreak (AutoRAN) and defense (RobustKV) methods against baselines (GCG, AutoDAN, PAIR, etc.) on open- and closed-source LLMs across various harmful benchmark datasets (AdvBench, HarmBench, StrongReject)

H&M Product Recommendations, the Kaggle Competition | Team Captain 2022.02 - 2022.04

• Built a recommendation system leveraging consumer profiles and purchase history; applied **Apriori** for product associations and **FunkSVD** for collaborative filtering. Replaced heuristic re-ranking with an **XGBoost** learning-to-rank model that integrated SVD embeddings, popularity signals, and user–item features, improving ranking quality and placing the team in the **top 10%** of the competition

Multi-dimensional Time-Series Classification & Prediction | Project Leader | 3 SCI Papers, 1 Conference Paper, 4 Patents 2021.09 - 2024.03

• Constructed a predictive maintenance system for multi-dimensional industrial data with **loss value, noise, and multiple patterns**. Developed methods include the **EM algorithm** for data imputation and denoising, an **attention-based network** for sensor selection, and a **customized LSTM** for prediction under multiple failure modes and operating conditions, finally achieving **5.58%** prediction error on aircraft engine datasets

SKILLS

• **Programming:** Python, SQL, C++, Java
• **ML/AI Frameworks:** PyTorch, HuggingFace, vLLM, GraphRAG, veRL, Ray, Megatron, FSDP, Deepspeed, wandb, OpenAI API, scikit-learn
• **Specialized Skills:** LLM Post-Training (LoRA, RLHF, DPO, PPO, GRPO, DAPO, GSPO), Adversarial ML, RAG, Prompt Engineering, Jailbreak Defenses, Model Steering, LLM Agent, Machine Learning, Software Engineering

AWARDS & SERVICE

Nisha and Vinod K. Singhi Graduate Fellowship (1 PhD student/year in CS Dept., $5,000), Stony Brook University	2024, 2025
Reviewer of NeurIPS 2025, IEEE CASE 2025&2024, IEEE TASE	2024, 2025
Teaching Assistant of CSE352 & CSE582 — Office hours, Grading, Student Communication	2024, 2025
Yangtze River Source Technology Innovation Scholarship (top 5% in Dept.), Shanghai Jiao Tong University	2023

SELECTED PUBLICATIONS

[1] **Yuhui Wang**, Changjiang Li, Guangke Chen, Jiacheng Liang, Ting Wang, "Reasoning or Retrieval? A Study of Answer Attribution on Large Reasoning Models", 2025 [Under Review at **ICLR** 2026]

[2] **Yuhui Wang**, Rongyi Zhu, Ting Wang, "Self-Destructive Language Model", 2025 [Under Review at **ICLR** 2026]

[3] Jiacheng Liang*, **Yuhui Wang***, Changjiang Li, Rongyi Zhu, Tanqiu Jiang, Neil Gong, Ting Wang, "GraphRAG under Fire", IEEE Symposium on Security and Privacy (**IEEE S&P**), 2026

[4] Tanqiu Jiang, Zian Wang, Jiacheng Liang, Changjiang Li, **Yuhui Wang**, Ting Wang, "RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction", International Conference on Learning Representations (**ICLR**), 2025 [10+ citations]

[5] **Yuhui Wang**, Andi Wang, Di Wang, Dong Wang, "Deep Learning-Based Sensor Selection for Multimodal Recognition and Prognostics Under Time-Varying Operating Conditions", IEEE Transactions on Automation Science and Engineering (**IEEE TASE**), 2025 [10+ citations]

[6] **Yuhui Wang**, Di Wang, "An Entropy-and Attention-Based Feature Extraction and Selection Network for Multi-Target Coupling Scenarios", IEEE International Conference on Automation Science and Engineering (**IEEE CASE**), 2023 [10+ citations]